QUIZ 1

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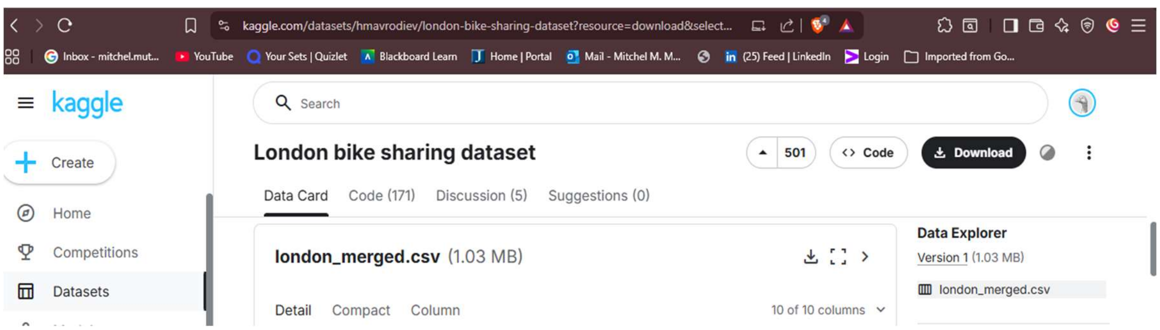
ID: 413

Data Source: London Bike

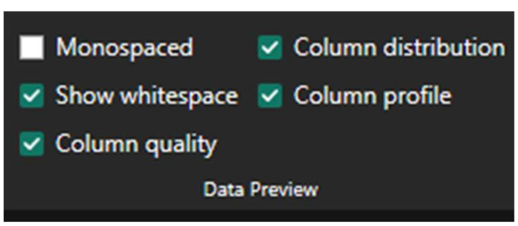
Link: https://www.kaggle.com/datasets/hmavrodiev/london-bike-sharing-dataset?resource=download&select=london_merged.csv

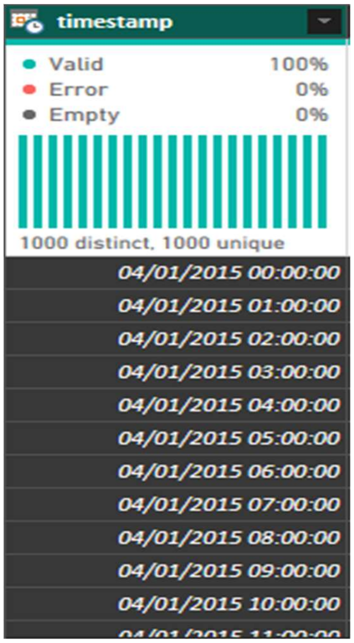
Dataset



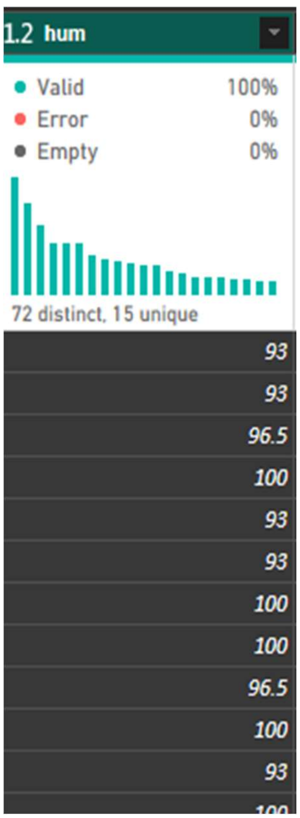
Section A

Q1



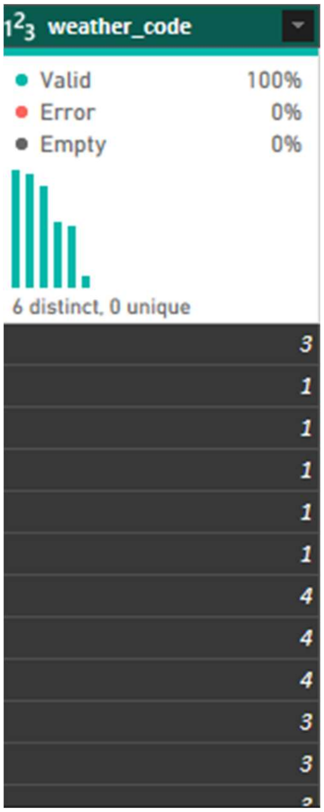


High cardinality (too many unique values) makes it hard to aggregate. Fix: Split Time and Date.



Percentage stored as a whole number (e.g., 80 instead of 0.8). Fix: Divide by 100.

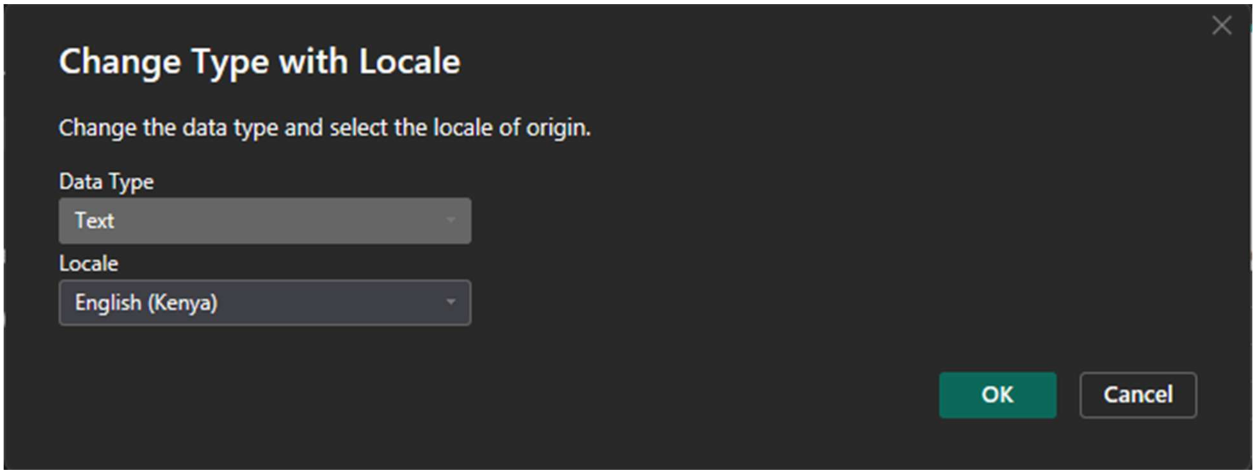
Numeric codes (1, 2, 3) are not descriptive. Fix: Merge with a look up.



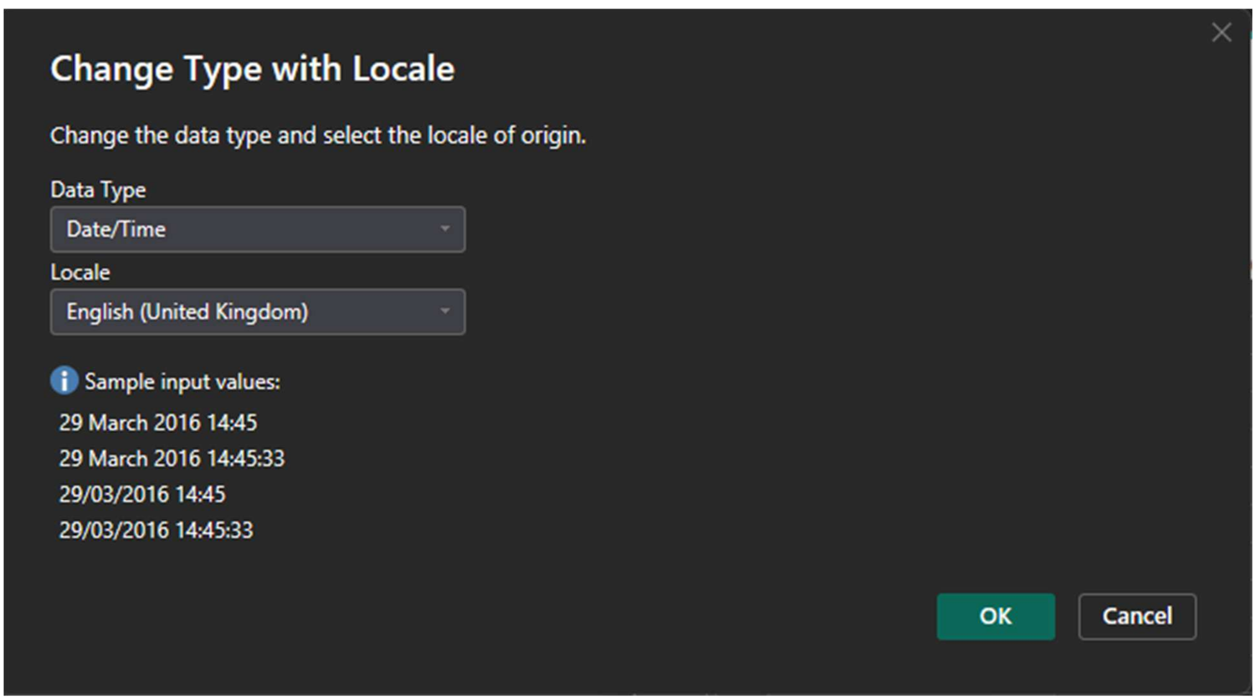
Non-descriptive names. Fix : Rename.

Q2

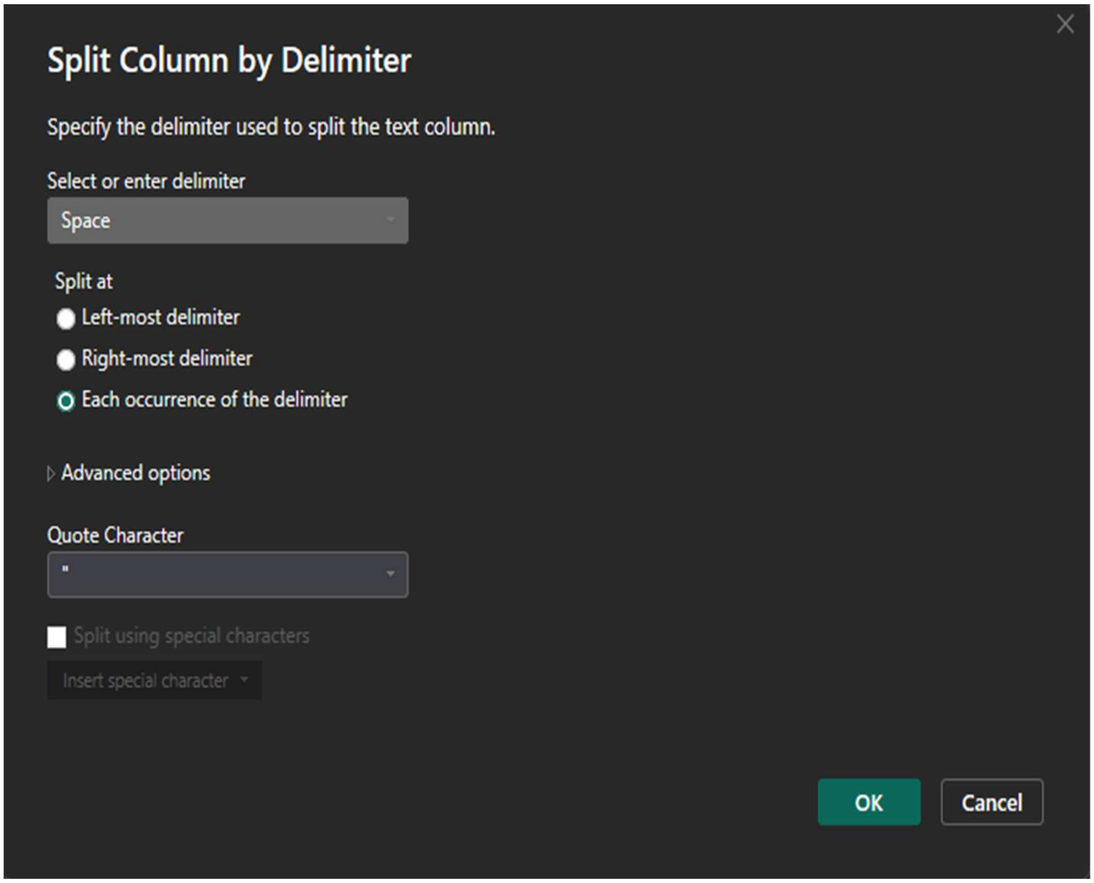
Before :

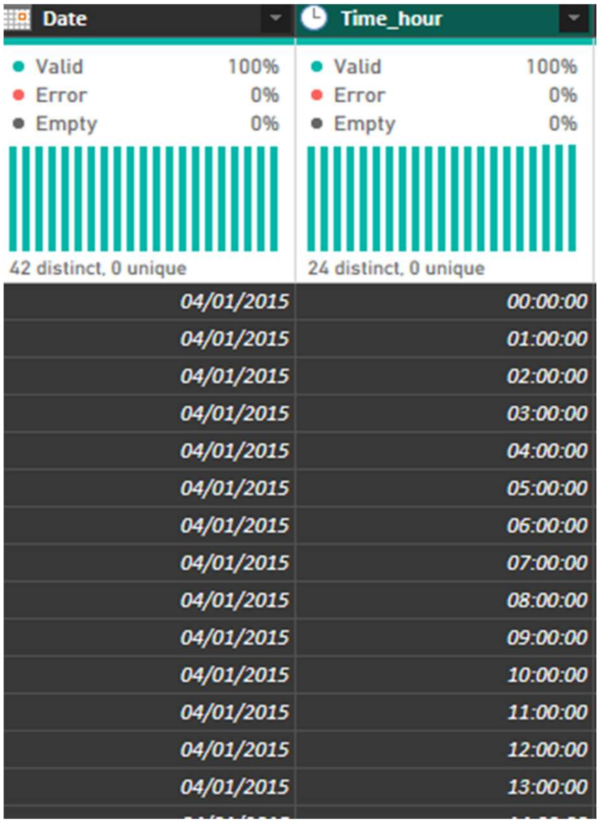


After :

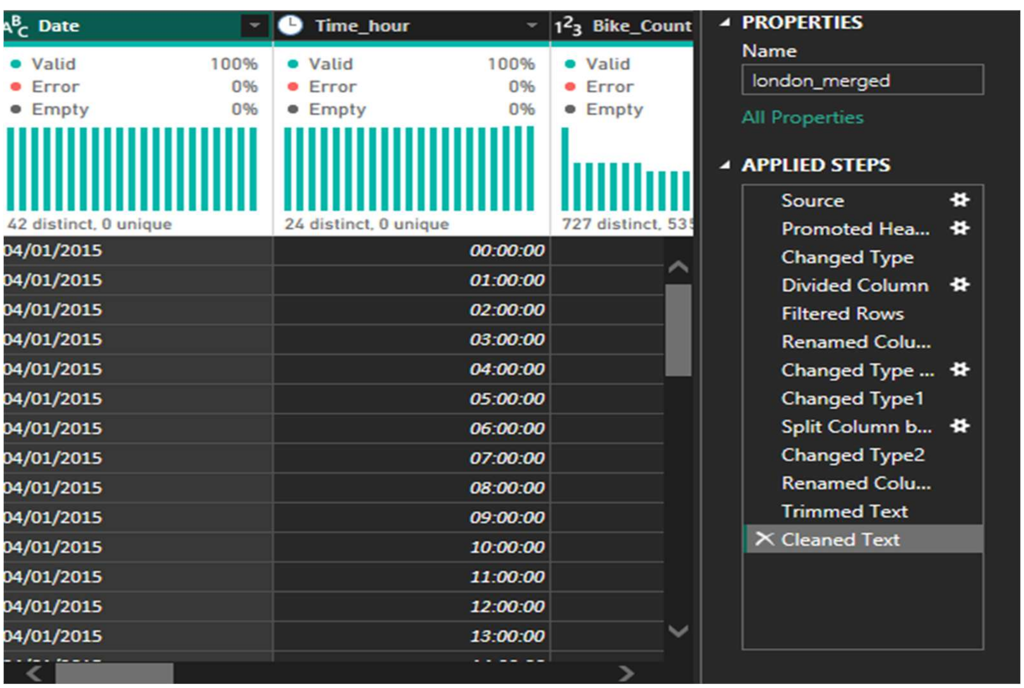


Q3

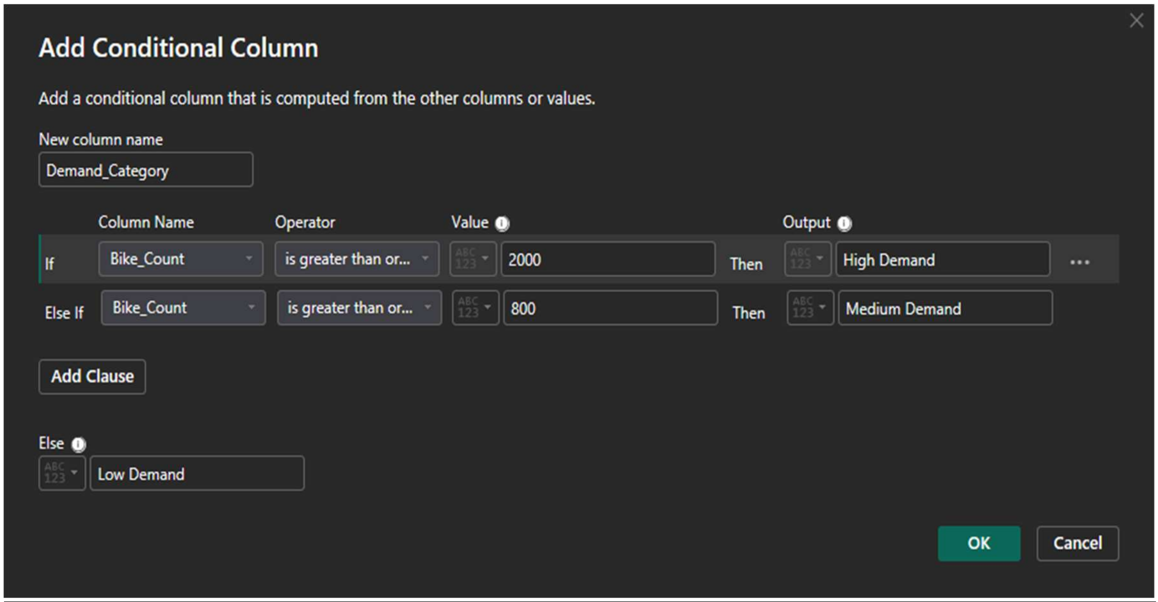

Split Column by Delimiter
Specify the delimiter used to split the text column.
Select or enter delimiter
Space
Split at
Left-most delimiter
Right-most delimiter
Each occurrence of the delimiter
Advanced options
Quote Character
"
Split using special characters
Insert special character
OK
Cancel

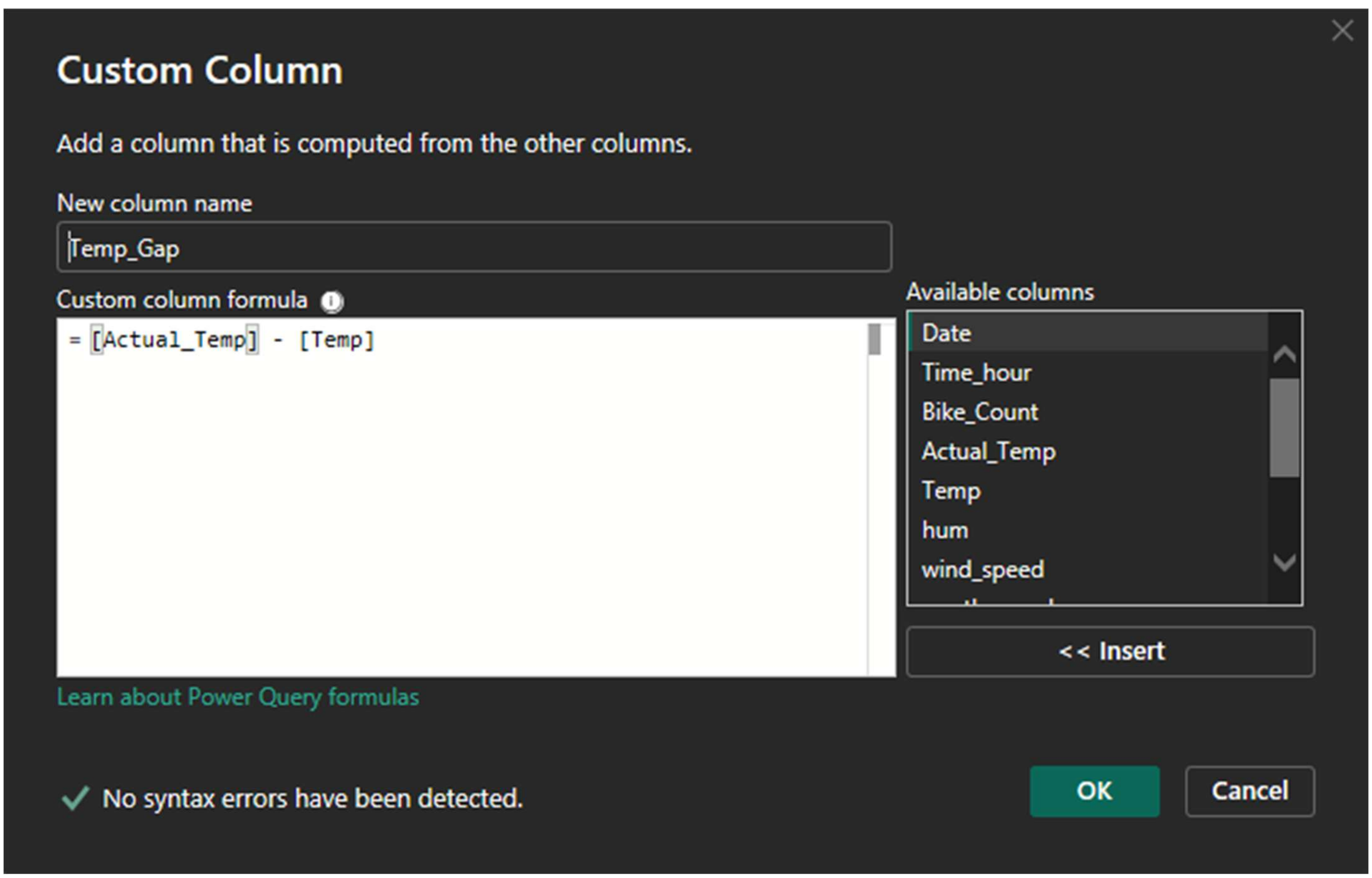


Trim and Clean :

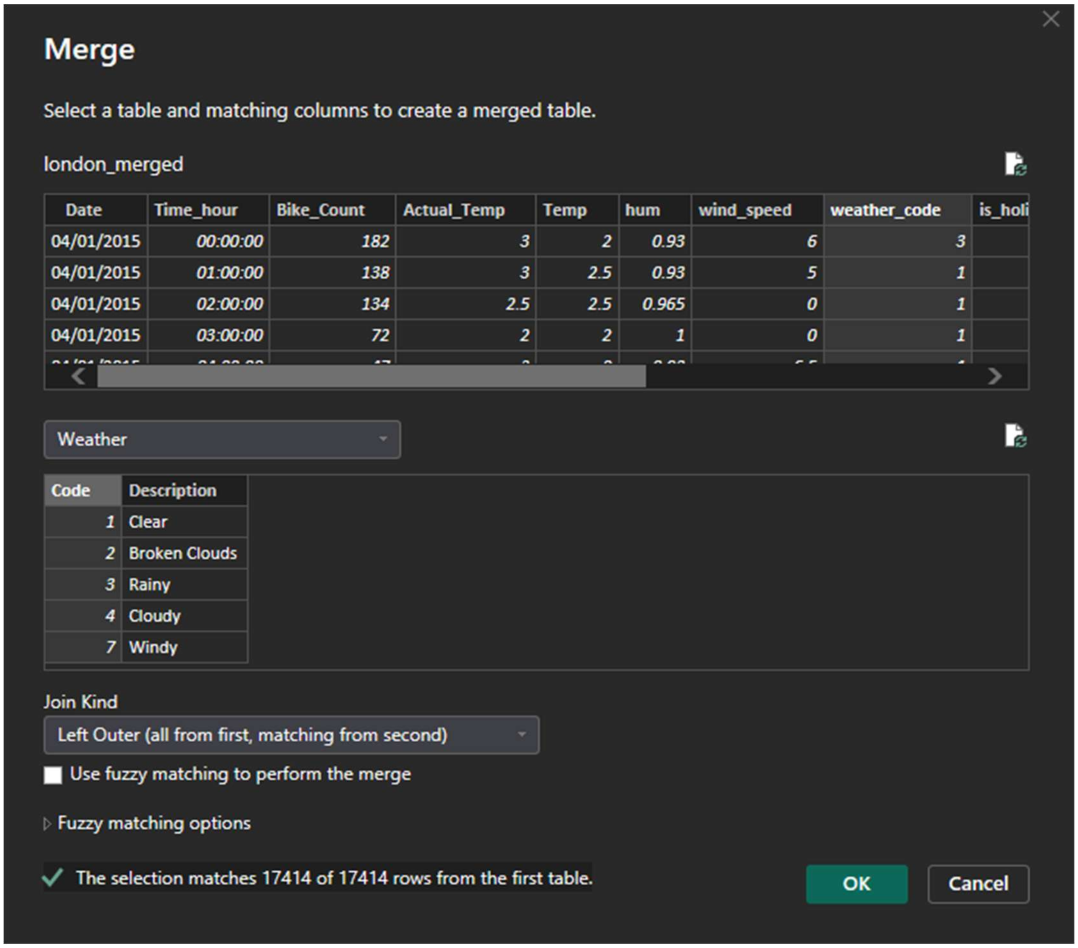


Q4

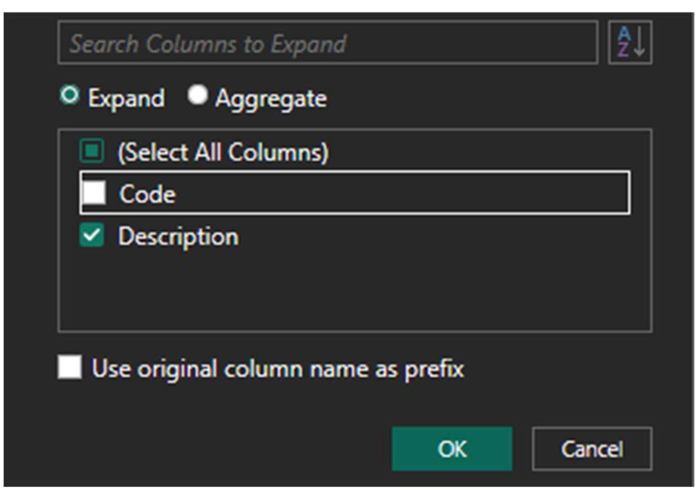

Add Conditional Column
Add a conditional column that is computed from the other columns or values.
New column name
Demand_Category
Column Name
Operator
Value
Output
If
Bike_Count
is greater than or...
2000
Then
High Demand
Else If
Bike_Count
is greater than or...
800
Then
Medium Demand
Add Clause
Else
Low Demand
OK
Cancel


Custom Column
Add a column that is computed from the other columns.
New column name
Temp_Gap
Custom column formula
= [Actual_Temp] - [Temp]
Available columns
Date
Time_hour
Bike_Count
Actual_Temp
Temp
hum
wind_speed
<< Insert
Learn about Power Query formulas
No syntax errors have been detected.
OK
Cancel

Q5



Merge
Select a table and matching columns to create a merged table.
london_merged
Date | Time_hour | Bike_Count | Actual_Temp | Temp | hum | wind_speed | weather_code | is_holi
04/01/2015 | 00:00:00 | 182 | 3 | 2 | 0.93 | 6 | 3
04/01/2015 | 01:00:00 | 138 | 3 | 2.5 | 0.93 | 5 | 1
04/01/2015 | 02:00:00 | 134 | 2.5 | 2.5 | 0.965 | 0 | 1
04/01/2015 | 03:00:00 | 72 | 2 | 2 | 1 | 0 | 1
Weather
Code | Description
1 | Clear
2 | Broken Clouds
3 | Rainy
4 | Cloudy
7 | Windy
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 17414 of 17414 rows from the first table.
OK
Cancel



Search Columns to Expand
Expand
Aggregate
(Select All Columns)
Code
Description
Use original column name as prefix
OK
Cancel

SECTION B

Q6

Star Schema Identification :

- **Fact Table:** London_Fact

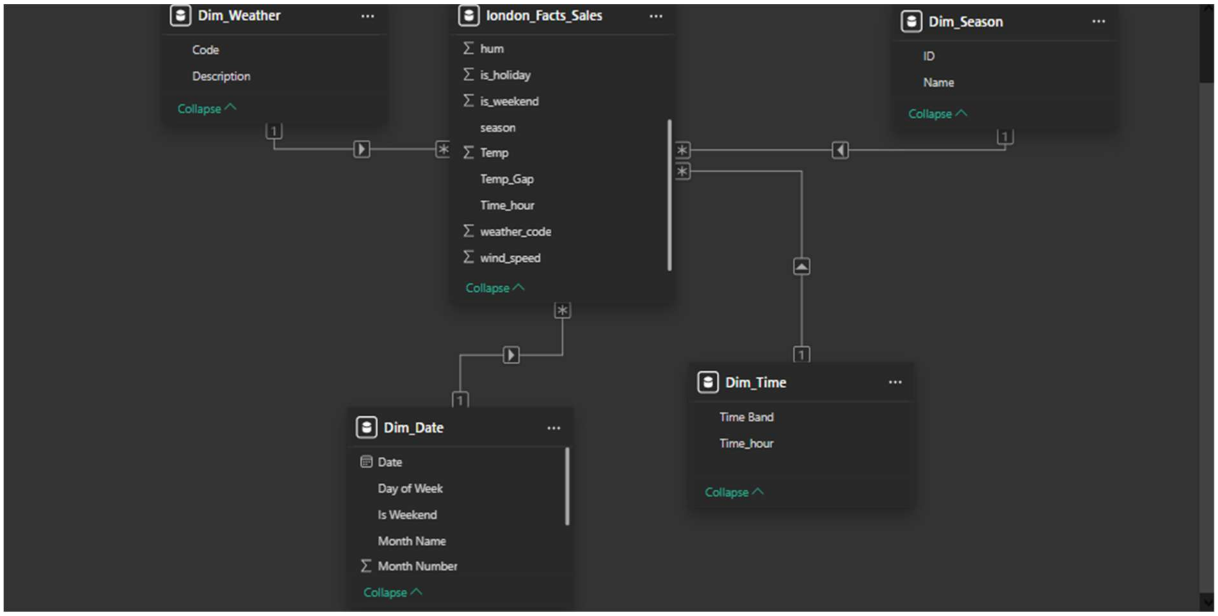
Justification: The quantitative observations (measures) of bike rentals and environmental data are included in this table. It links to every lookup table and has the greatest grain.

- **Dimension Table:** Dim_Weather (Mapping Table)

Justification: The descriptive qualities (such as "Clear" and "Rainy") in this table provide the fact table's numerical codes context.

- **Dimension Table:** Dim_Date (Calendar Table)
- **Dimension Table:** Dim_Season (Season Table)
- **Dimension Table:** Dim_Hour (Hours Table)

Q7



Relationship 1: Dim_Weather to Fact[weather_code]

- Cardinality: One-to-Many
- Cross Filter Direction: Single
- Status: Active
- Why correct: Each numeric weather code appears once in the lookup table (Dimension) but can be recorded multiple times (Many) in the rental logs (Fact).

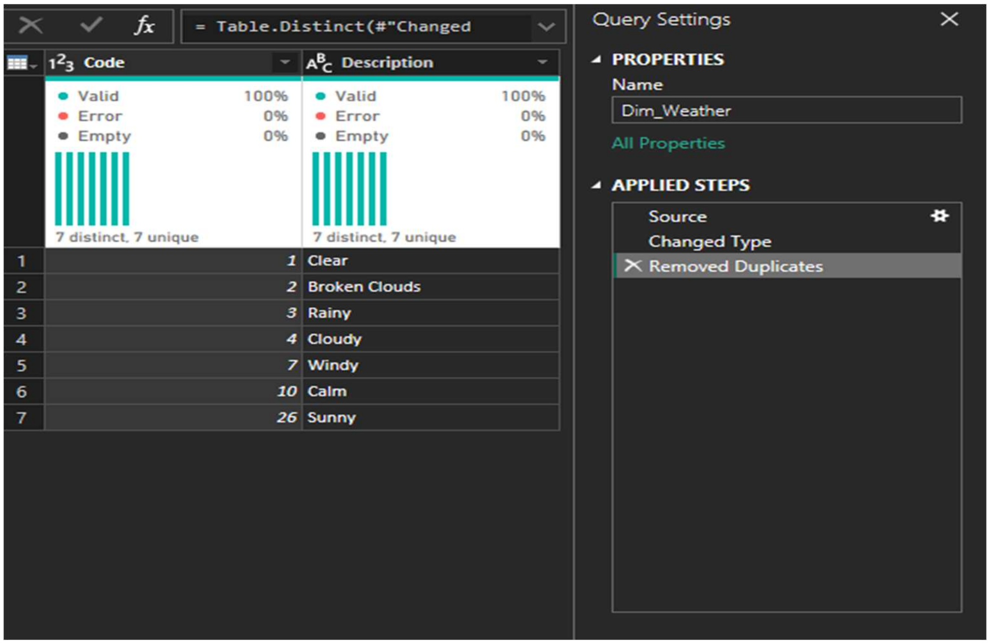
Relationship 2: Dim_Time [Hour] to Fact [Time_Hour]

- Cardinality: One-to-Many
- Cross Filter Direction: Single
- Status: Active
- Why correct: This allows us to slice thousands of rental records by a specific hour (e.g., 8:00 AM) to analyze peak commute patterns.

Identify One modelling mistake:

- **Mistake:** Connecting Dim_Season directly to Dim_Weather, snow only happens in winter.
- **Impact:** This creates a Snowflake Schema rather than a Star Schema. While logically true, it increases complexity and can slow down report performance in Power BI compared to having both tables connect directly to the Fact table.

Q8



Verification Method: In Power Query, I selected the Code column in Dim_Weather, went to the View tab, and checked Column Distribution. It confirmed "100% Unique."

Action Taken: Even though it was unique, I applied the **"Remove Duplicates"** step on the primary key column to ensure that if the source file is updated with redundant rows in the future, the model will not break with a "Many-to-Many" error.

Q9

- **Decision: Yes,** a data table must exist.
- **Reasons:**
  i) **Reporting Flexibility:** It allows us to group data by attributes not found in the raw data, such as "Fiscal Year," "Month Short Name," or "Day of the Week.
  ii) Allows for **Year-over-year comparisons** of Bike Rentals**.**
  iii) **Time Intelligence:** Without a dedicated Date table, Power BI cannot reliably perform DAX functions like SAMEPERIODLASTYEAR or TOTALYTD because the timestamp in the London data is hourly, not a continuous daily sequence.
- **Columns to include**: Date, Month Name, Year.

Bonus: Small Dictionary

Table	Column	Meaning	Type	Key/Attribute
London_Fact	Bike_Count	Count of new bike shares	Whole Number	Measure
London_Fact	weather_code	Numeric ID for weather	Whole Number	Foreign Key
Dim_Date	Date	Unique calendar date	Date	Primary Key
Dim_Date	Is Weekend	Flag for Saturday/Sunday	Text	Attribute
Dim_Weather	Description	Clear, Rainy, Snow, etc.	Text	Attribute
Dim_Time	Hour	24-hour clock (0-23)	Whole Number	Primary Key
Dim_Time	Time Band	Period (e.g., Rush Hour)	Text	Attribute
Dim_Season	Season Name	Spring, Summer, etc.	Text	Attribute

VISUALIZATION